

From Narrow to General: Can Self-Learning Systems Lead to AGI?

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Abstract

This paper argues that today's large language models (LLMs) are unlikely to reach artificial general intelligence (AGI) through scale alone, because their knowledge is frozen after training. We propose that AGI requires a **dynamic self-learning system**: one that updates itself while acting and learns from its own mistakes, as humans do, rather than being re-trained or fine-tuned by people.

We examine three questions. First, can narrow AI (ANI) serve as the foundation for AGI? Second, does data quality matter more than data quantity? Third, how could we prove a system has become AGI? We outline experiments at Grow Labs using low-parameter models to test whether small systems can generalise to things they were never trained on and improve from their own errors.

1. Introduction

AI today is powerful but narrow. Whether it can become general is still an open question.

Term	Meaning	Example
ANI (Artificial Narrow Intelligence)	Excellent at one task or domain	Chess engines, image classifiers, translation models
AGI (Artificial General Intelligence)	Learns and performs across most tasks a human can, including new ones	Not yet achieved
ASI (Artificial Superintelligence)	Surpasses the best humans in almost every domain	Theoretical

Research questions

- 1 Can an LLM become AGI, or even go beyond it?
- 2 Can narrow AI be the foundation for AGI?
- 3 Does the quality of training data matter for general intelligence?
- 4 Is a self-learning system, not a fine-tuned one, required for AGI?
- 5 When a system becomes AGI, how do we prove it?

2. How current AI works

An LLM is trained to predict the next token (a word or part of a word) given the text before it. It is extremely good at this prediction.

$$P(w_1, \dots, w_n) = \prod_{t=1..n} P(w_t | w_1, \dots, w_{t-1})$$

The model learns this probability from huge amounts of text. After training, its weights are fixed. When you talk to it, it does not change; it only uses what it already learned.

Does the model know what it is doing? Our view is that it does not, in the human sense. It has no goals of its own and no awareness of its task. Some researchers disagree. They argue that to predict text well, a

model must build internal models of the world, and that this is a form of understanding. This debate is still open, and this paper does not claim to settle it.

What is clear is this: prediction alone, with frozen weights, is different from how humans learn. A human changes after every mistake. A deployed LLM does not.

3. Can LLMs become AGI?

LLMs may be a necessary part of AGI, but on their own they are probably not enough.

Arguments for	Arguments against
Scaling laws: performance improves predictably with more parameters, data and compute (Kaplan et al., 2020; Hoffmann et al., 2022)	Frozen weights: the model cannot learn after deployment
New abilities appear at larger scales, such as multi-step reasoning (Wei et al., 2022)	Some "emergent" abilities may be a result of how we measure them (Schaeffer et al., 2023)
One model handles code, maths, language and images	Hallucination: confident but false answers
With tools and memory, LLMs already act as agents	No lasting memory of past experience
	Weak at truly new problems it has never seen (Chollet, 2019)

Our position: an LLM is a strong **foundation**, like a brain's knowledge store. AGI likely needs an extra layer on top: continuous learning from experience. This is the focus of the next section.

4. Dynamic Self-Learning Systems

Our core proposal is that AGI needs a system that learns while it works, not one that waits for humans to re-train it.

	Fine-tuning (today)	Dynamic self-learning (proposed)
Who starts learning	Humans collect data and re-train	The system itself, during use
When it learns	Offline, in separate training runs	Continuously, while acting
Source of learning	Curated datasets	Its own actions, results and mistakes
Speed of update	Weeks or months	Seconds to hours
Human analogy	A student who only learns in exams	A person who learns from every mistake

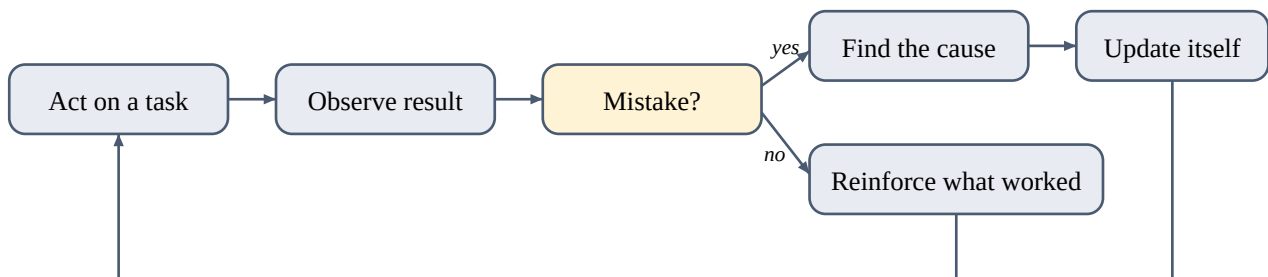


Figure 1. The self-learning loop: act, check, correct, repeat.

Main challenges

- **Catastrophic forgetting:** when a neural network learns something new, it often overwrites old knowledge (Kirkpatrick et al., 2017).
- **Learning the wrong lesson:** a system that updates itself can reinforce errors or be manipulated.
- **Knowing it made a mistake:** a system needs a reliable signal of what is correct. In chess this is easy (win or lose). In real life it is often unclear.
- **Safety:** a system that changes itself is harder to test and control.

Related work includes continual learning, reinforcement learning, and self-improving methods such as STaR, where a model learns from its own correct reasoning (Zelikman et al., 2022).

5. How ANI helps build AGI

Narrow AI can help build AGI in three ways: as parts, as tools, and as lessons.

- 1 **As parts.** Many specialist models (vision, language, planning, memory) could be combined under one controller. This is similar to how the human brain has specialised regions.
- 2 **As tools.** A narrow model can help build the next model, for example by writing code, cleaning data or checking answers.
- 3 **As lessons.** Narrow systems show us what learning methods work, which we can then generalise.

The chess example. A chess engine can beat any human champion. But it did not learn every possible game. Chess has roughly 10^{120} possible games (Shannon, 1950), far more than atoms in the observable universe. No computer can store them all.

Instead, systems like AlphaZero learned by playing against themselves and building a *general sense* of which positions are good (Silver et al., 2018). They then handle positions they have never seen. This is important for our thesis: even narrow AI **generalises** beyond its training data, and it learned from its own mistakes through self-play. The open question is whether this kind of learning can work outside a closed game with clear rules.

Can a 1T-parameter narrow model build AGI? Suppose we train a 1-trillion-parameter model on one domain only: the task of building AI. Could it design an AGI? This is currently unanswered. A narrow model may lack the broad knowledge that general reasoning needs. But it might speed up AI research enough to matter. We list this as an open question, not a claim.

6. Does data matter?

Yes. Existing evidence suggests data quality matters a lot, but it may matter less for a self-learning system.

- **Quantity:** Chinchilla showed many models were under-trained. For a fixed budget, more data and fewer parameters worked better (Hoffmann et al., 2022).
- **Quality:** small models trained on carefully chosen, textbook-quality data can match much larger models on some tasks (Gunasekar et al., 2023).
- **Self-generated data:** AlphaZero used no human games at all. It created its own data through self-play (Silver et al., 2018).

This suggests a hypothesis for our work: **a self-learning system depends less on the starting dataset, because it creates new data from its own experience.** If true, the quality of its feedback signal (knowing right from wrong) matters more than the size of its first dataset.

7. Beyond AGI

If a system can truly improve itself, it may not stop at human level. That is why self-learning is both the path to AGI and the path beyond it.

- **Intelligence explosion:** Good (1965) argued that a machine smart enough to improve itself could trigger rapid, repeated self-improvement.
- **Speed and scale:** even at human-level ability, an AI can run faster, copy itself and never sleep. This alone could make it more capable than any human.
- **Self-modifying systems:** Schmidhuber's Gödel Machine (2003) is a theoretical design for a system that rewrites its own code when it can prove the change is better.

This also raises safety concerns. A system that improves itself must stay aligned with human goals at every step. Any research on dynamic self-learning should include safety limits from the start.

8. How do we prove it is AGI?

No single test can prove AGI today. We propose that AGI should be judged on how well a system **learns new things**, not on how much it already knows.

Test	What it measures	Weakness
Turing Test (Turing, 1950)	Can it pass as human in conversation?	Measures imitation, not intelligence
Benchmarks (exams, coding tests)	Knowledge and skill on fixed tasks	Can be memorised from training data
ARC (Chollet, 2019)	Solving new visual puzzles from a few examples	Covers one narrow type of reasoning
Levels of AGI (Morris et al., 2023)	Performance and generality compared with skilled humans	A framework, not a single test
Economic test	Can it do most jobs humans are paid for?	Hard to measure, depends on society

Proposed criteria for AGI. A system should be called AGI only if it can:

- 1 **Transfer:** solve tasks in domains it was never trained on.
- 2 **Learn quickly:** reach good performance on a new task from few examples, like a human.
- 3 **Learn from mistakes:** measurably improve after its own errors, without human re-training.
- 4 **Remember:** keep old skills while learning new ones (no catastrophic forgetting).
- 5 **Stay consistent:** show these abilities across many domains, not just one.

Criteria 3 and 4 are what current LLMs lack. They are also what our experiments aim to test.

9. Proposed experiments at Grow Labs

We test our ideas on small models first, because small models are cheap to train and easy to study. Our current model has about 20 million parameters, trained on 10,000 samples on an Apple M1 Max. We plan to scale to 12 billion parameters and, later, toward 1 trillion.

Experiment	Question	How we measure it
E1. Unseen knowledge	Can the model answer questions about things not in its training data?	Accuracy on a held-out test set with new topics

Experiment	Question	How we measure it
E2. Parallel prediction	Can it predict several outputs or tasks at the same time?	Accuracy on each task, run together vs separately
E3. Learning from mistakes	Does it improve after its own errors, without human re-training?	Error rate before and after a self-correction loop
E4. Forgetting	Does new learning erase old knowledge?	Score on old tasks after learning new ones
E5. Data quality	Does a small clean dataset beat a larger noisy one?	Same model, two datasets, compare test scores

Limitations. Results from a 20M-parameter model may not hold at larger scales. Some abilities only appear in big models. We will repeat key experiments at each scale step to check this.

10. Conclusion and open questions

LLMs are excellent predictors and a strong foundation, but frozen prediction is probably not enough for AGI. We argue the missing piece is dynamic self-learning: a system that improves from its own mistakes while it works. Narrow AI already shows this is possible in closed worlds like chess. The challenge is making it work in the open, messy real world.

Open questions

- Can self-learning work without a clear right-or-wrong signal?
- Can a very large narrow model help design a general one?
- How do we stop a self-learning system from forgetting or learning the wrong lessons?
- How do we keep a self-improving system safe?

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